

PRST Class (29/10/2020)

$$X \sim N(\mu, \sigma^2)$$

$$M(t) = E(e^{tX})$$

$$f(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}}$$

$$E(e^{tX}) = \int_{-\infty}^{\infty} e^{tx} \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{1}{2}\frac{(x-\mu)^2}{\sigma^2}} dx$$

$$\begin{aligned} &= \int_{-\infty}^{\infty} e^{(u+\sigma z)t} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz \quad \begin{array}{l} \frac{x-\mu}{\sigma} = z \\ dx = \sigma dz \\ x = \mu + \sigma z \end{array} \\ &= e^{ut} \int_{-\infty}^{\infty} e^{\sigma t z} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}z^2} dz \quad \begin{array}{l} \sigma t z - \frac{1}{2}z^2 \\ = -\frac{1}{2}(z - \sigma t)^2 \\ \times e^{\frac{1}{2}\sigma^2 t^2} \end{array} \\ &= e^{ut + \frac{\sigma^2 t^2}{2}} \int_{-\infty}^{\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z - \sigma t)^2} dz \\ &= e^{ut + \frac{\sigma^2 t^2}{2}} \times 1 \\ &= e^{ut + \frac{\sigma^2 t^2}{2}} \end{aligned}$$

$$\boxed{\frac{\ln\left(\frac{S}{n} - E(X)\right)}{\sqrt{\text{Var}(X)}}} \quad (7)$$

Eg. The height dist of a group of 10,000 people is normal with height 64.5" and sd 4.5".
Find the number of people whose height is
(a) less than 63" but greater than 55.5,
(b) greater than 73.5.

Sol: X : height of an individual in the group.
 $X \sim N(64.5, (4.5)^2)$

$$\begin{aligned} &P(55.5 < X < 63) \\ &= P\left(\frac{55.5 - 64.5}{4.5} < \frac{X - 64.5}{4.5} < \frac{63 - 64.5}{4.5}\right) \quad \begin{array}{l} \text{Let } Z = \frac{X - \mu}{\sigma} \sim N(0,1) \\ \text{Then } \frac{X - \mu}{\sigma} \sim N(0,1) \end{array} \\ &= P(-2 < Z < 1) \quad \begin{array}{l} P(a < X < b) \\ \Phi(1) - \Phi(-2) \end{array} \\ &= \Phi(1) - \Phi(-2) \\ &= 0.82 \quad \begin{array}{l} \Phi(1) = 0.841 \\ \Phi(-2) = 0.0227 \end{array} \end{aligned}$$

$$\begin{aligned} \text{ans} &= 10,000 \times 0.82 \\ &= 8,200 \end{aligned}$$

Eg. Mean of a normal dist is 50 and the values are greater than 60.

Eg. Mean of a normal
5% of the values are greater than
Find the sd of the dist.

Sol: $\mu = 50$

$$P(X > 60) = \frac{5}{100} = 0.05 \quad \left[\frac{\Phi(1.64)}{\Phi(1.64)} = 0.95 \right]$$

$$\Rightarrow P\left(\frac{X-50}{\sigma} > \frac{10}{\sigma}\right) = 0.05 \quad Z = \frac{X-\mu}{\sigma}$$

$$\Rightarrow P\left(Z > \frac{10}{\sigma}\right) = 0.05$$

$$\Rightarrow P\left(Z \leq \frac{10}{\sigma}\right) = 1 - 0.05 = 0.95$$

$$\Rightarrow \Phi\left(\frac{10}{\sigma}\right) = 0.95$$

$$\Rightarrow \frac{10}{\sigma} = 1.64$$

Eg. Assuming that height dist of a group of
people is normal. Find ~~mean~~ mean
and sd if 84% of the people
have height less than 65.2" and
68% have height between 65.2" and
62.8"

Sol: $X \sim N(\mu, \sigma^2)$

$$P(X < 65.2) = .89 \quad \left[\frac{\Phi(1)}{\Phi(1)} = 0.16 \right] \quad Z = \frac{X-\mu}{\sigma}$$

$$P(62.8 < X < 65.2) = 0.168 \quad \text{--- (1)}$$

and $\Rightarrow P\left(\frac{X-\mu}{\sigma} < \frac{65.2-\mu}{\sigma}\right) = .89$

$$P\left(\frac{X-\mu}{\sigma} < \frac{65.2-\mu}{\sigma}\right) = 0.84 \quad \left(Z \sim N(0,1) \right) \quad \text{--- (ia)}$$

$$P\left(\frac{X-\mu}{\sigma} < \frac{65.2-\mu}{\sigma}\right) = 0.84 \quad \text{(ia)}$$

$$\Rightarrow P\left(\frac{62.8-\mu}{\sigma} < \frac{X-\mu}{\sigma} < \frac{65.2-\mu}{\sigma}\right) = 0.68$$

$$\Rightarrow P\left(\frac{62.8-\mu}{\sigma} < Z < \frac{65.2-\mu}{\sigma}\right) = 0.68 \quad \text{(iia)}$$

$$\Phi\left(\frac{65.2-\mu}{\sigma}\right) = 0.84$$

$$\Phi\left(\frac{65.2-\mu}{\sigma}\right) - \Phi\left(\frac{62.8-\mu}{\sigma}\right) = 0.68$$

$$\Phi\left(\frac{62.8-\mu}{\sigma}\right) = 0.84 - 0.68 = 0.16 \quad \left| \Phi(-1) = 0.16 \right.$$

$$\frac{62.8-\mu}{\sigma} = -1 \quad \text{--- (b)}$$

$$\begin{aligned} \Phi\left(\frac{65.2-\mu}{\sigma}\right) &= 0.84 = 1 - 0.16 \\ &= 1 - \Phi(-1) \\ &= \Phi(1) \end{aligned}$$

$$\frac{65.2-\mu}{\sigma} = 1 \quad \text{--- (c)}$$

Ex. first moment about 10 is 40.
Fourth moment about 50 is 48.) Normal $N(\mu, \sigma)$

— σ

Ex. first moment about 10 is 40.
Fourth moment about 50 is 48.) Normal
Find μ, σ . $N(\mu, \sigma)$